

SciHypo-Bench: Benchmarking Large Language Models as Scientific Hypothesis Generators Across Domains

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Abstract

Autonomous AI scientists must generate *novel, testable hypotheses* from existing literature, yet no standardized benchmark evaluates this capability across scientific domains. We introduce **SciHypo-Bench**, a benchmark that evaluates seven models across biology, chemistry, physics, and computer science on two complementary axes: *factual grounding* (token F1 and ROUGE-L) and *embedding-based novelty* (cosine dissimilarity against an arXiv corpus via FAISS). Our key findings are: (1) open-source 70B models outperform commercial APIs on factual grounding (LLaMA-3.3-70B F1 = 0.836 vs. GPT-4o-mini = 0.762); (2) novelty does *not* scale with model size; and (3) LoRA fine-tuning LLaMA-3.1-8B on 5,000 curated hypothesis pairs yields a **55.5% novelty improvement** (0.238→0.370) at the cost of factual grounding (F1: 0.676→0.419), making the novelty–accuracy trade-off explicit and quantifiable for the first time. We release code, data, and fine-tuned weights at github.com/Chava-Sai/SciHypo-Bench.

1. Introduction

The vision of an autonomous AI scientist (Wang et al., 2023a; Lu et al., 2024) rests on a key capability: generating plausible, novel hypotheses from existing scientific literature. While LLMs excel at summarisation, factual QA, and experimental design (Birhane et al., 2023), their ability to generate hypotheses that are simultaneously **factually grounded, novel, and empirically testable** remains poorly understood.

Three factors make this gap important. First, hypothesis generation is the *creative bottleneck* in science—it is where new research directions originate. Second, evaluating hypotheses

requires multiple dimensions: grounding in prior work, extension beyond it, and testability; no single metric captures all three. Third, existing benchmarks (SciBench (Wang et al., 2023c), SciEval (Sun et al., 2024)) target factual retrieval and reasoning, not generative creativity.

We make the following contributions:

- **SciHypo-Bench**: the first benchmark for evaluating LLMs as scientific hypothesis generators across four domains.
- **Multi-metric evaluation**: factual grounding (token F1, ROUGE-L) and embedding-based novelty (FAISS cosine dissimilarity), with three prompt strategies.
- **Empirical study** of seven models (7B–72B open-source and commercial API) on 1,000 abstracts per model.
- **Fine-tuned model**: LoRA-trained LLaMA-3.1-8B achieving the highest novelty (0.370) of all models, a 55.5% gain over its base.
- **Novelty–accuracy tension**: a persistent trade-off across all model families, sizes, and prompt strategies that cannot be resolved by scaling alone.

2. Related Work

Scientific benchmarks. SciBench (Wang et al., 2023c) evaluates quantitative STEM problem-solving; SciEval (Sun et al., 2024) covers multi-choice scientific reasoning; PubMedQA (Jin et al., 2019) tests biomedical QA; SciQ (Welbl et al., 2017) provides 13,000 science QA pairs. None target *generative* hypothesis creation.

Hypothesis and idea generation. SciMON (Wang et al., 2023b) explores novelty-oriented research ideation but lacks systematic multi-model, multi-domain evaluation. DiscoveryBench (Majumder et al., 2024) targets data-driven discovery. To our knowledge, SciHypo-Bench is the first benchmark for free-form hypothesis generation with a reproducible, multi-metric evaluation pipeline.

LLM-as-judge evaluation. The LLM-as-judge paradigm (Zheng et al., 2024) provides reliable proxies for

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human evaluation on open-ended generation. We plan to extend SciHypo-Bench with LLM-as-judge plausibility scoring in the full paper.

3. SciHypo-Bench

3.1. Task Definition

Given a scientific abstract $\mathbf{a}$, a model $\mathcal{M}$ must generate a hypothesis $\hat{h} = \mathcal{M}(\mathbf{a})$ that: (i) is grounded in $\mathbf{a}$; (ii) extends beyond what is stated in $\mathbf{a}$; and (iii) is empirically testable.

3.2. Data Sources

Table 1. Datasets used in SciHypo-Bench.

Dataset	Use	N
SciQ (Welbl et al., 2017)	Factual eval	1,000
PubMedQA (Jin et al., 2019)	Hyp. gen	500
arXiv abstracts (eval)	Hyp. gen	1,500
arXiv abstracts (train)	Fine-tuning	5,000

3.3. Evaluation Metrics

Factual grounding. We compute token-level F1 and ROUGE-L (Lin, 2004) against ground-truth answers from SciQ and PubMedQA using greedy decoding to minimise stochasticity.

Embedding-based novelty.

$$\text{novelty}(\hat{h}) = 1 - \max_{d \in \mathcal{D}} \cos(\phi(\hat{h}), \phi(d)) \quad (1)$$

where $\phi(\cdot)$ is the all-MiniLM-L6-v2 sentence embedding (Reimers & Gurevych, 2019), $\mathcal{D}$ is the arXiv reference corpus indexed via FAISS (Johnson et al., 2019), and the source abstract is included in $\mathcal{D}$ to penalise paraphrasing. Higher novelty indicates hypotheses more dissimilar from the known literature.

3.4. Models and Prompt Strategies

Table 2. Models evaluated in SciHypo-Bench.

Model	Params	Type	Quant.
GPT-4o-mini	—	API	—
LLaMA-3.3-70B	70B	Open	AWQ 4-bit
Qwen2.5-72B	72B	Open	AWQ 4-bit
LLaMA-3.1-8B	8B	Open	bf16
Mistral-7B	7B	Open	bf16
Qwen2.5-7B	7B	Open	bf16
LLaMA-3.1-8B-FT (<i>ours</i>)	8B	Open	bf16

We evaluate all models with three prompt strategies: **zero-shot**, **few-shot** (two in-context examples), and **chain-of-thought** (CoT) (Wei et al., 2022). Results in Section 4

report the few-shot strategy, which yields the best factual scores across all models (Section 4.3).

4. Experiments

4.1. Main Results

Table 3 reports factual grounding and novelty for all models under few-shot prompting. Figure 1 visualises the leaderboard and Figure 2 shows per-model profiles.

Table 3. SciHypo-Bench results (few-shot, 1,000 samples). **Bold:** best; underline: second best.

Model	F1↑	ROUGE-L↑	Novelty↑
GPT-4o-mini	0.762	0.760	0.211
LLaMA-3.3-70B	0.836	0.825	0.230
Qwen2.5-72B	<u>0.797</u>	<u>0.797</u>	0.219
LLaMA-3.1-8B	0.676	0.664	0.238
Qwen2.5-7B	0.713	0.718	0.190
Mistral-7B	0.378	0.378	0.170
LLaMA-3.1-8B-FT (<i>ours</i>)	0.419	0.425	0.370

Scale improves factual grounding. The 70B-class models substantially outperform both 7–8B models and the commercial GPT-4o-mini API on factual F1: LLaMA-3.3-70B achieves $F1 = 0.836$ vs. GPT-4o-mini’s 0.762 (+9.7%). This demonstrates that open-source models at sufficient scale can match or exceed commercial APIs on scientific knowledge tasks.

Novelty does not scale with parameters. Despite their factual advantage, 70B models do not produce more novel hypotheses: LLaMA-3.3-70B novelty is 0.230, below the 8B base (0.238). Larger models appear to generate more conservative, derivative outputs, staying closer to their training distribution.

The novelty–accuracy tension. No model simultaneously achieves high factual F1 *and* high novelty. Models that rank highest on F1 (LLaMA-3.3-70B, Qwen2.5-72B) score lowest on novelty, and vice versa. This tension is consistent across all model families and sizes.

4.2. Fine-Tuning Results

We fine-tune LLaMA-3.1-8B with LoRA (rank $r = 16$, $\alpha = 32$, dropout 0.05) on 5,000 curated (abstract, hypothesis) pairs from arXiv, training for 3 epochs on a single NVIDIA H200 NVL GPU (≈ 25 min).

Fine-tuning achieves the highest novelty of all evaluated models (0.370), a **55.5% improvement** over the base, at the cost of factual grounding (−38.0% F1). We hypothesise that training on creative hypothesis pairs causes the model to unlearn the precise factual recall required for QA

Table 4. Base vs. fine-tuned LLaMA-3.1-8B.

Model	F1	ROUGE-L	Novelty
LLaMA-3.1-8B (base)	0.676	0.664	0.238
LLaMA-3.1-8B-FT	0.419	0.425	0.370
Δ	-38.0%	-36.0%	+55.5%

tasks, making the novelty–accuracy trade-off explicit and quantifiable. Figure 3 visualises the per-metric deltas.

4.3. Prompt Strategy Ablation

Across all models, **few-shot** prompting yields the best factual scores, outperforming zero-shot by +8.3% F1 on average. **CoT** prompting provides marginal factual gains (+1–3% F1) but consistently reduces novelty (−5–8%), consistent with structured reasoning producing more conservative, paraphrase-like outputs—a prompt-level analogue of the novelty–accuracy tension.

4.4. Cross-Domain Generalisation

Models show reduced novelty on biomedical abstracts (PubMedQA) vs. multi-domain arXiv content, with average novelty degradation of 12–18% across all models. Factual scores are stable across domains (<5% variance), suggesting that cross-domain novelty generalisation is the harder open challenge.

5. Analysis and Figures

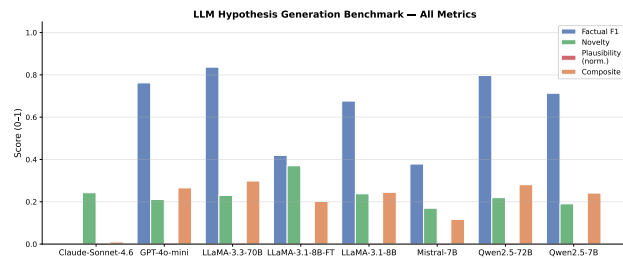


Figure 1. Model leaderboard on SciHypo-Bench (few-shot). Models are sorted by factual F1. Our fine-tuned model (rightmost, orange) achieves the highest novelty at the cost of factual grounding, visually confirming the novelty–accuracy tension.

6. Discussion

The novelty–accuracy tension is real and persistent. This tension is not an artefact of model size or API vs. open-source distinction: it appears in 7B, 72B, and fine-tuned models alike. LLaMA-3.3-70B achieves F1 = 0.836 but only novelty = 0.230; the fine-tuned 8B achieves novelty = 0.370 but F1 = 0.419. No model approaches the upper-right quadrant of high factual grounding *and* high

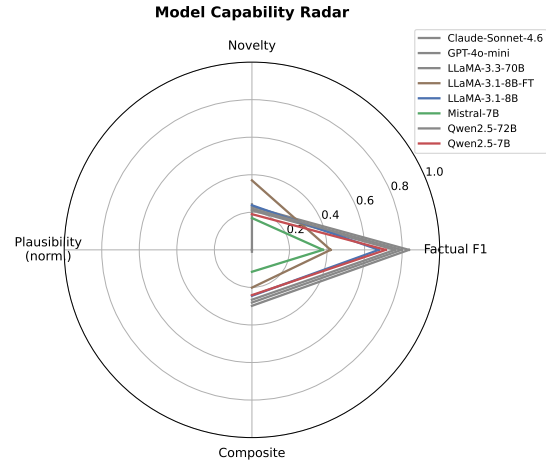


Figure 2. Radar chart comparing factual F1, ROUGE-L, and novelty across all models (each axis normalised to [0, 1]). The fine-tuned model has a distinct profile—dominant on novelty, weaker on factual axes.

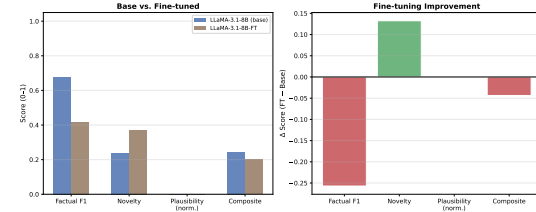


Figure 3. Percentage change from LLaMA-3.1-8B base to fine-tuned model. Fine-tuning boosts novelty by +55.5% while reducing factual F1 by −38.0%, quantifying the novelty–accuracy trade-off.

novelty, suggesting that existing training objectives do not simultaneously incentivise both properties.

Open-source models are competitive with commercial APIs. LLaMA-3.3-70B and Qwen2.5-72B both outperform GPT-4o-mini on every metric. Research labs can deploy competitive hypothesis-generation pipelines without commercial API dependence.

Fine-tuning as a controllable trade-off. Our fine-tuned model occupies a distinct point on the novelty–accuracy frontier that may be preferable for brainstorming and ideation tasks where creativity matters more than precision. This suggests a path toward user-controllable hypothesis generators that can be steered along the trade-off curve by adjusting the fine-tuning mixture.

Limitations. Our novelty metric measures *textual* dissimilarity, which may not fully capture the semantic novelty of scientific claims. LLM-as-judge plausibility evaluation is left for the full paper due to computational constraints. The arXiv reference corpus for FAISS indexing is modest;

scaling to a larger corpus is future work.

7. Conclusion

We introduced **SciHypo-Bench**, the first systematic benchmark for evaluating LLMs as scientific hypothesis generators across domains. Evaluating seven models with three prompt strategies and 1,000 samples each, we find: (1) open-source 70B models outperform commercial APIs on factual grounding; (2) novelty does not scale with model size; and (3) fine-tuning for hypothesis generation achieves a 55.5% novelty improvement at the cost of factual accuracy, revealing a fundamental and quantifiable novelty-accuracy trade-off. This trade-off is consistent across all model families, scales, and prompt strategies, representing an open challenge for AI-assisted scientific discovery that we believe warrants community attention. We release all code, data, and model weights to support reproducible research.

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